Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Klastoets 3
VRAESTEL

Class Test 3
QUESTION PAPER

Module EBN111
Elektrisiteit en Elektronika
May 2017

Module EBN111
Electricity and Electronics
May 2017

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***	MEMO		
Maksimum punte: *Maximum marks:*	16	Volpunte: *Full marks:*	15
Duur van vraestel: *Duration of paper:*	30 min + 10 min to complete OCR form *30 min + 10 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		8 + OCR	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
   All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	
	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session.
Do step 1 in the next 30 minutes of the session.
Do step 2 within the following 5 minutes.

STEP 0 (0 min to 5 min): Complete the front page of the OCR form

STEP 1 (5 min to 35 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (35 min to 40 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.
- The Assessment ID is: **2017EBN111C3**
- Use a **mechanical pencil** to complete the form so that you can erase errors.
- **Do not use commas!** Please make use of full stops.
- When writing a complex number, **place the *j* at the front of the imaginary number!**

Examples

01	–	1	.	4	5					m	A							
02	3	4	0	+	j	2	0			V								
03	1	5	.	2	–	j	1	6										
04	–	j	1	3						Ω								

INSTRUKSIES

Doen stap 0 in die eerste 5 minute van die sessie.
Doen stap 1 in die volgende 30 minute van die sessie.
Doen stap 2 binne die daaropvolgende 5 minute.

STAP 0 (0 min to 5 min): Voltooi die voorblad van die OKH vorm

STAP 1 (5 min to 35 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (35 min to 40 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.
- Die "Assessment ID" is **2017EBN111C3**
- Gebruik **'n meganiese potlood** om antwoorde in te vul sodat jy foute kan uitvee.
- **Moenie kommas gebruik nie!** Maak eerde gebruik van punte.
- **Plaas die *j* aan die begin van die imaginêre getal** wanneer jy komplekse getalle skryf!

Voorbeelde

01	–	1	.	4	5					m	A							
02	3	4	0	+	j	2	0			V								
03	1	5	.	2	–	j	1	6										
04	–	j	1	3						Ω								

Questions 01-04 [4]

Given two sinusoid signals v_1 and v_2, where

$$v_1 = -4\sin(20t - 60°),$$
$$v_2 = 4\cos(20t + 150°).$$

a) Determine $v = v_1 + v_2$, and write the final answer in the form $A cos(\omega t + \phi)$ where $A > 0$ and $-180° \leq \phi \leq 180°$.

b) Which statement is correct?

1. v_1 leads v_2 2. v_2 leads v_1

c) What is the phase angle ϕ_d between v_1 and v_2? $(0 \leq \phi_d \leq 180°)$

01	A
02	ϕ
03	Choose 1 or 2.
04	ϕ_d

Vraag 01-04 [4]

Twee sinusiedseine v_1 en v_2 word gegee:

$$v_1 = -4\sin(20t - 60°),$$
$$v_2 = 4\cos(20t + 150°).$$

a) Bepaal $v = v_1 + v_2$, en skryf die finale antwoord in die vorm $A cos(\omega t + \phi)$ met $A > 0$ en $-180° \leq \phi \leq 180°$.

b) Watter stelling is korrek?

2. v_1 loop voor v_2 2. v_2 loop voor v_1

c) Wat is die fasehoek ϕ_d tussen v_1 en v_2? $(0 \leq \phi_d \leq 180°)$

01	A
02	ϕ
03	Kies 1 of 2.
04	ϕ_d

a) $V_1 = 4\cos(20t + 30°)$

$v_2 = 4\cos(20t + 150°)$

$\bar{V}_1 = 4\angle 30°, \quad \bar{V}_2 = 4\angle 150°$

$V = \bar{V}_1 + \bar{V}_2 = 4\angle 30° + 4\angle 150° = 4\angle 90°$

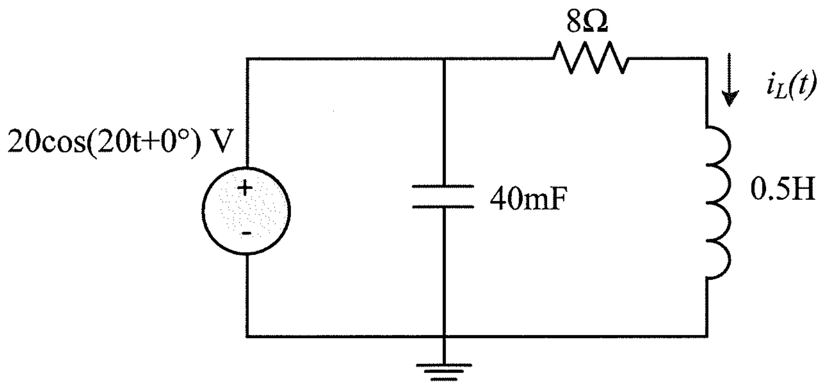
$V = 4\cos(20t + 90°)$

$\Rightarrow \underline{A = 4, \quad \phi = 90°}$

b) $\underline{2}$

c) $\phi_d = 150° - 30° = \underline{120°}$

Question 05-06 [2]		Vraag 05-06 [2]	
Determine $i_L(t)$. (The final answer must be in the following form $Acos(\omega t + \theta)$, $A > 0$, and $\|\theta\| \leq 180°$.		Bepaal $i_L(t)$. (Die finale antwoord moet in die volgende vorm wees: $Acos(\omega t + \theta)$, $A > 0$, en $\|\theta\| \leq 180°$.	
05	A	**05**	A
06	θ	**06**	θ

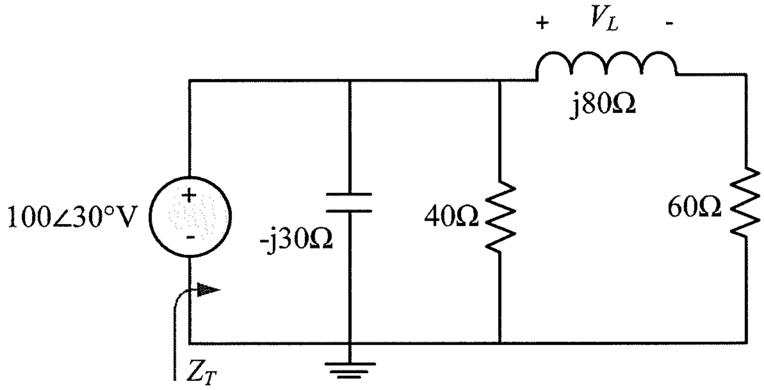


$\bar{V} = 20\angle 0°$.

$Z_L = j\omega L = j \times 20 \times 0.5\,\Omega = j10\Omega$

$\bar{I}_L = \dfrac{\bar{V}}{8+j10} = \dfrac{20\angle 0°}{12.81\angle 51.34°} = 1.5617\angle -51.340°$

Question 07 & 08 [4]		Vraag 07 & 08 [4]	
Find Z_T and V_L in **rectangular form.**		Bepaal Z_T en V_L in **reghoekige vorm.**	
07	Z_T	**07**	Z_T
08	V_L	**08**	V_L



$$Z_T = (-j30) \| 40 \| (60+j80)$$

$$= \frac{-j30 \times 40}{40-j30} \| (60+j80)$$

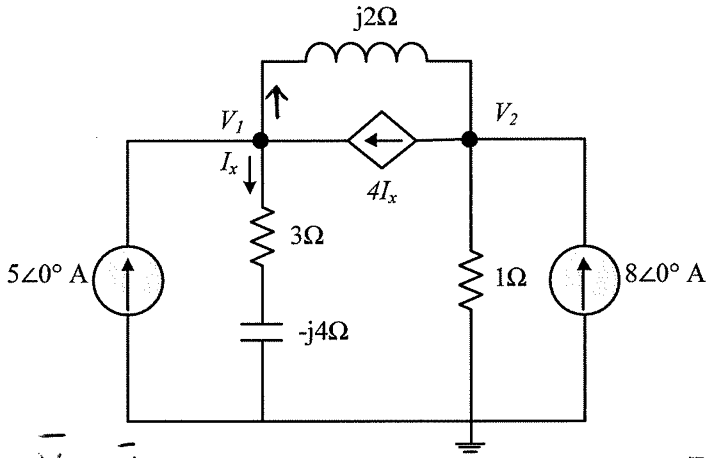
$$= 24\angle -53.13 \| 100\angle 53.13$$

$$= \frac{2400}{24\angle -53.13 + 100\angle 53.13} = \frac{2400}{74.4+j60.8} = \underline{19.341 - j15.806\ (\Omega)}$$

$$\bar{V}_L = 100\angle 30° \cdot \frac{j80}{60+j80} = \frac{8000\angle 120°}{100\angle 53.13°} = 80\angle 66.870°\ V$$

$$= \underline{31.425 + j73.570\ (V)}$$

Question 09 & 12 [4]		Vraag 09 & 12 [2]	
Use nodal analysis to set up two equations of the following form: $aV_1 + bV_2 = 5\angle 0°$ $cV_1 + dV_2 = 8\angle 0°$ Give the coefficients *a*, *b*, *c*, and *d* in **rectangular form.**		Gebruik node analise om twee vergelykings van die volgende form op te stel: $aV_1 + bV_2 = 5\angle 0°$ $cV_1 + dV_2 = 8\angle 0°$ Gee die koeffisiënte in **reghoekige vorm.**	
09	*a*	**09**	*a*
10	*b*	**10**	*b*
11	*c*	**11**	*c*
12	*d*	**12**	*d*



$$V_1: \quad \frac{\bar{V}_1}{3-j4} + \frac{\bar{V}_1 - \bar{V}_2}{j2} = 4I_x + 5\angle 0°, \quad I_x = \frac{\bar{V}_1}{3-j4} \qquad (1)$$

$$V_2: \quad \frac{\bar{V}_1 - \bar{V}_2}{j2} + 8\angle 0° = 4I_x + \frac{V_2}{1} \qquad (2)$$

$$(1) \Rightarrow \frac{\bar{V}_1 - \bar{V}_2}{j2} - \frac{3\bar{V}_1}{3-j4} = 5\angle 0°, \quad \left(\frac{1}{j2} - \frac{3}{3-j4}\right)\bar{V}_1 + \left(-\frac{1}{j2}\right)\bar{V}_2 = 5\angle 0°$$

$$\Rightarrow \underline{a = -0.36 - j0.98}, \quad \underline{b = j0.5}$$

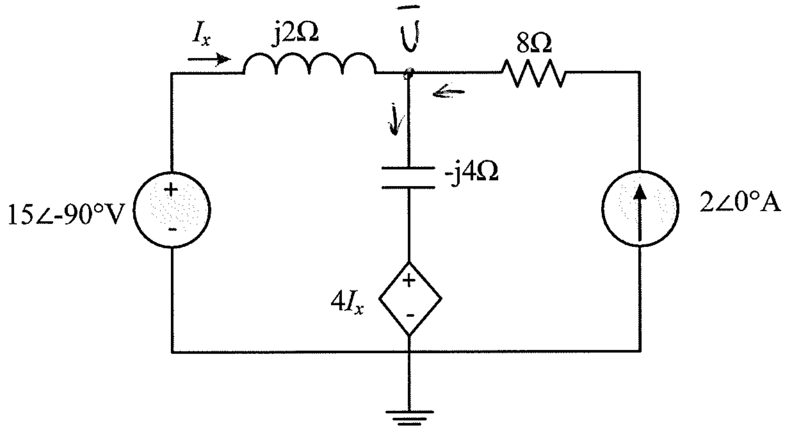
$$(2) \Rightarrow \frac{4\bar{V}_1}{3-j4} + \bar{V}_2 - \frac{\bar{V}_1 - \bar{V}_2}{j2} = 8\angle 0°$$

$$\left(\frac{4}{3-j4} - \frac{1}{j2}\right)\bar{V}_1 + \left(1 + \frac{1}{j2}\right)\bar{V}_2 = 8\angle 0°$$

$$\Rightarrow \underline{c = 0.48 + j1.14}$$

$$\underline{d = 1 + \frac{1}{j2} = 1 - j0.5}$$

Questions 13 [2]		Vraag 13 [2]	
What is the individual contribution of the current source to $\bar{I}_x$? Give the final answer in **rectangular form**.		Wat is die individuele bydra van die stroombron tot $\bar{I}_x$? Gee die finale antwoord in **reghoekige vorm**.	
13	$\bar{I}_{x(A)}$	**13**	$\bar{I}_{x(A)}$



Let voltage soure as 0, set nodal equation at $\bar{V}$

$$I_x + 2\angle 0^\circ = \frac{\bar{V} - 4I_x}{-j4}, \quad I_x = -\frac{\bar{V}}{j2}$$

$$\Rightarrow -\frac{\bar{V}}{j2} + 2\angle 0^\circ = \frac{\bar{V} + \frac{4\bar{V}}{j2}}{-j4} \quad ①$$

$$① \times (-j4) \Rightarrow 2\bar{V} - j8\angle 0^\circ = \bar{V} - j2\bar{V}$$

$$\Rightarrow (1+j2)\bar{V} = j8\angle 0^\circ, \quad \bar{V} = \frac{j8\angle 0^\circ}{1+j2} = 3.2 + j1.6 \text{ (V)}$$

$$\bar{I}_{x(A)} = -\frac{\bar{V}}{j2} = -\frac{3.2+j1.6}{j2} = -0.8 + j1.6 \text{ (A)}$$

PRACTICE PAGE / OEFENBLAD

Student number													
Assessment ID	2	0	1	7	E	B	N	1	1	1	C	3	

Remember: Write a complex number as $a \pm \boldsymbol{j}\, b$ where $\boldsymbol{j}$ is at the **start** of the imaginary number!

	Answer	Unit
01	$A =$ 4	
02	$\phi =$ 90	°
03	2	
04	$\phi_a =$ 120	°
05	$A =$ 1.5617	A
06	$\theta =$ −51.340	°
07	$Z_T =$ 19.341 − j15.806	Ω
08	$V_L =$ 31.425 + j73.570	V
09	$a =$ −0.36 − j0.98	
10	$b =$ j0.5	
11	$c =$ 0.48 + j1.14	
12	$d =$ 1 − j0.5	
13	$\bar{I}_{x(A)} =$ −0.8 + j1.6	A

To assist us with training data, please do the following on the OCR form:
In row 14, fill in: $20\angle 10$
In row 15, fill in: $4.56\angle 12.4$
These are not for marks.